Counted Cantor against the cost model's 10 g² multiplications

2026-10-14. Frozen protocol `PROTOCOL.md` (sha256 `c27e54a3c7bbbb33…`). Native instrument `examples/jacobian_op_count.rs`. All 10400 counted results agree with the repository's group law (0 mismatches).

Result (exact computation)

Mean counts for one Cantor addition of two random degree-`g` divisors over `F_1009`, 200 pairs per genus, with the total weighting an inversion as 8 multiplications. `ratio` is `log2` of the total over the model's `10 g²`; `local e` is the exponent between this genus and the previous measured one.

g	mults	inversions	total	10 g²	ratio (bits)	local e
2	124.6	5.00	164.7	40	2.041	—
4	505.5	8.01	569.6	160	1.832	1.807
6	1266.4	11.06	1354.9	360	1.912	2.022
8	2508.6	14.04	2620.9	640	2.034	2.294
10	4448.1	17.18	4585.6	1000	2.197	2.507
14	10657.0	23.32	10843.5	1960	2.468	2.624
16	15132.6	26.36	15343.5	2560	2.583	2.600
24	46026.4	38.63	46335.5	5760	3.008	2.766
32	101876.9	50.70	102282.5	10240	3.320	2.752

Verdict (fixed wording). The model is below textbook Cantor by 2.041 bits at genus 2 and 3.320 bits at genus 32 (addition, `p = 1009`; doubling 3.210 bits and `p = 65521` 3.231 bits at genus 32). The ratio is outside the one-bit band at every measured genus. The fitted exponent is 2.347 (95% bootstrap interval 2.346 to 2.349) for addition, 2.349 (2.348 to 2.350) for doubling and 2.319 (2.319 to 2.320) for addition at `p = 65521`: the exponent 2 is outside each interval. The intervals cover divisor sampling only; the local exponents rise from about 1.8 to about 2.75, so a single power law is an approximation.

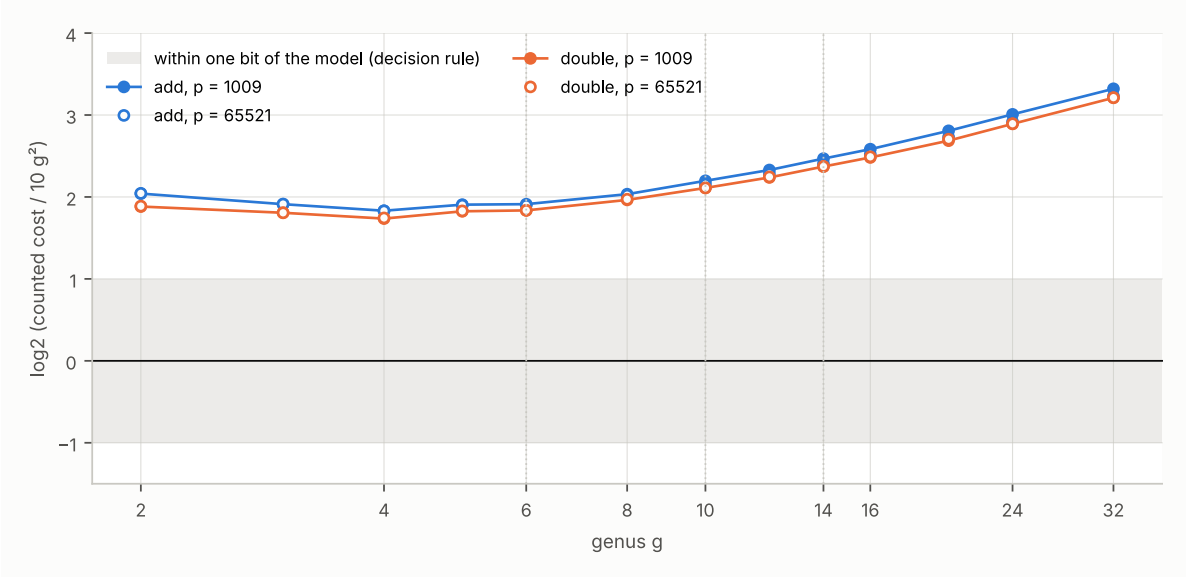


Figure 1: Counted cost over `10 g²` by genus, addition and doubling, two primes. The grey band is the one-bit decision rule; dotted lines mark genus 6, 10 and 14, where most extension routes sit. Source: `summary.csv`.

Could the gap move a verdict? (exact count on the committed JSON)

Of the 848 `formula_advantage` verdicts, 14 lie within 1 bit of the 8-bit margin, 23 within 2 bits and 32 within 3 bits; 50 more are `formula_marginal`. If the measured cost replaced `10 g²` and the **whole** route cost scaled with it, 24 of the 848 would drop below the margin. That is an upper bound on fragility: routes priced by a different solver do not scale entirely with the group operation.

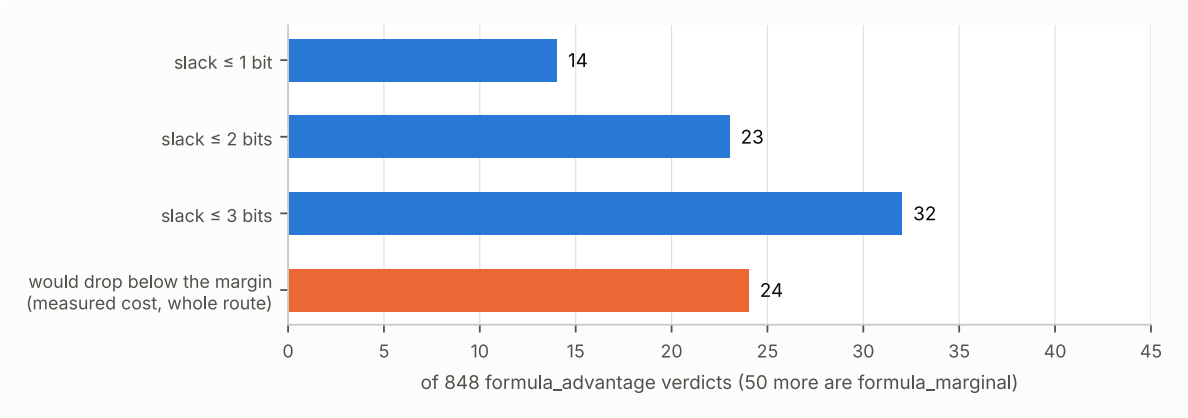


Figure 2: Advantage verdicts close to the margin, and the conservative count that the measured cost would push below it. Source: `summary.csv`.

Claims and scope

- **Exact computation:** the counts per genus, the fit, and the verdict counts above.
- **Not shown:** that the model is wrong. Textbook Cantor with extended Euclid is not the cheapest algorithm (explicit formulas exist for `g = 2, 3`), so the ratio is a statement about this algorithm, not about the best one. The model's unit may also count an optimised operation.
- **Not shown:** anything about `F_p` of cryptographic size; the primes are `1009` and `65521`, and the two primes differ by at most 0.09 bits in the ratio at any measured genus, so no dependence on `p` is visible here.
- **Dominant caveat:** 846 of the 848 advantage verdicts are extension-field `cover_descent` routes marked "proposed", whose genus-k cover over `GF(p)` is not established (the route's own note). Whether those covers exist matters more to those verdicts than any constant in this report; see the odd-characteristic descent study.